

page mode cycle, subject to the same conditions as in normal operation (previously described). These operations can be intermixed in consecutive page mode cycles and performed in any order. The maximum number of consecutive page mode cycles is limited by t_{RAS} . Page mode operation is ended when $\overline{RAS}$ transitions to inactive, coincident with or following $\overline{CAS}$ inactive transition.

REFRESH CYCLES

The dynamic RAM design is based on capacitor charge storage for each bit in the array. This charge degrades with time and temperature, thus each bit must be periodically **refreshed** (recharged) to maintain the correct bit state. Bits in the MCM511000A require refresh every 8 milliseconds while refresh time for the MCM51L1000A is 64 milliseconds.

Refresh is accomplished by cycling through the 512 row addresses in sequence within the specified refresh time. All the bits on a row are refreshed simultaneously when the row is addressed. Distributed refresh implies a row refresh every 15.6 microseconds for the MCM511000A and 124.8 microseconds for the MCM51L1000A. Burst refresh, a refresh of all 512 rows consecutively, must be performed every 8 milliseconds on the MCM511000A and 64 milliseconds on the MCM51L1000A.

A normal read, write, or read-write operation to the RAM will refresh all the bits (2048) associated with the particular row decoded. Three other methods of refresh, **$\overline{RAS}$ -only refresh**, **$\overline{CAS}$ before $\overline{RAS}$ refresh**, and **Hidden refresh** are available on this device for greater system flexibility.

$\overline{RAS}$ -Only Refresh

$\overline{RAS}$ -only refresh consists of $\overline{RAS}$ transition to active, latching the row address to be refreshed, while $\overline{CAS}$ remains high (V_{IH}) throughout the cycle. An external counter is employed to ensure all rows are refreshed within the specified limit.

$\overline{CAS}$ Before $\overline{RAS}$ Refresh

$\overline{CAS}$ before $\overline{RAS}$ refresh is enabled by bringing $\overline{CAS}$ active before $\overline{RAS}$. This clock order activates an internal

refresh counter that generates the row address to be refreshed. External address lines are ignored during the automatic refresh cycle. The output buffer remains at the same state it was in during the previous cycle (hidden refresh).

Hidden Refresh

Hidden refresh allows refresh cycles to occur while maintaining valid data at the output pin. Holding $\overline{CAS}$ active at the end of a read or write cycle, while $\overline{RAS}$ cycles inactive for t_{RP} and back to active, starts the hidden refresh. This is essentially the execution of a $\overline{CAS}$ before $\overline{RAS}$ refresh from a cycle in progress (see Figure 1).

$\overline{CAS}$ BEFORE $\overline{RAS}$ REFRESH COUNTER TEST

The internal refresh counter of this device can be tested with a **$\overline{CAS}$ before $\overline{RAS}$ refresh counter test**. This test is performed with a read-write operation. During the test, the internal refresh counter generates the row address, while the external address supplies the column address. The entire array is refreshed after 512 cycles, as indicated by the check data written in each row. See **$\overline{CAS}$ before $\overline{RAS}$ refresh counter test cycle** timing diagram.

The test can be performed after a minimum of **eight $\overline{CAS}$ before $\overline{RAS}$** initialization cycles. Test procedure:

1. Write "0"s into all memory cells with normal write mode.
2. Select a column address, read "0" out and write "1" into the cell by performing the **$\overline{CAS}$ before $\overline{RAS}$ refresh counter test, read-write cycle**. Repeat this operation 512 times.
3. Read the "1"s which were written in step 2 in normal read mode.
4. Using the same starting column address as in step 2, read "1" out and write "0" into the cell by performing the **$\overline{CAS}$ before $\overline{RAS}$ refresh counter test, read-write cycle**. Repeat this operation 512 times.
5. Read "0"s which were written in step 4 in normal read mode.
6. Repeat steps 1 to 5 using complement data.